TIR-Net: Task Integration Based on Rotated Convolution Kernel for Oriented Object Detection in Aerial Images

Hao Li, Rong Pan, Gang Liu, Min Dang, *Graduate Student Member, IEEE*, Qijie Xu, Xu Wang, and Bo Wan

***Abstract*— The application of oriented object detection in the field of aerial images has gained substantial attention and made significant progress. However, most one-stage object detectors struggle to extract rotation-invariant features of oriented objects using ordinary convolutions. And the structure of two parallel vision subtasks can result in the inconsistency between the classification and regression. In this article, we propose a Task Integration based on a Rotated convolution kernel Network (TIR-Net) consisting of three modules: selective rotation of the kernel (SRK), regression feature refinement (RFR), and task integration (TI). Specifically, an SRK module enhances classification features by applying rotated convolution kernels selectively, introducing rotational invariance to the features. An RFR module places more emphasis on feature extraction of large aspect ratio objects to improve their perceptibility. A TI module integrates classification and regression features to alleviate the inconsistency between classification and regression. Experimental evaluations on the benchmarks for oriented object detection indicate that our method achieves excellent detection performance.**

***Index Terms*— Aerial images, misaligned tasks, one-stage detector, oriented object detection, rotation-invariant.**

I. Introduction

OBJECT detection in aerial images has been a significant research topic, aiming to locate and classify objects of interest within images. It finds application in diverse scenarios such as land planning [1], [2], [3], [4], vegetation monitoring [5], [6], military reconnaissance [7], and marine resource surveys [8]. The objects in aerial images exhibit arbitrary orientations, because of their acquisition from an aerial perspective. Besides, prominent scale variations and intricate backgrounds further increase the difficulty of oriented object detection. Recently, many rotated object detectors have applied different technical means to improve detection performance [9], [10], [11], [12], [13], [14].

Most advanced methods for oriented object detection in aerial images have greatly benefited from the R-CNN frameworks [15], [16], [17], [18], [19]. These methods predict bounding boxes as regions of interest (RoIs) first and then leverage the features of RoIs for category recognition. These methods make the extracted features more detailed and accurate, culminating in high accuracy but high computational complexity and slow inference speeds. In contrast with R-CNN-based detectors, one-stage detectors are widely used due to their high efficiency and simple design [20], [21], [22]. One-stage detectors do not need the first stage for predicting the proposals and directly predict the classification and regression results of objects end-to-end. However, one-stage methods build classification and regression subnetworks using two parallel branches independently for detection. The classification task aims to familiarize peculiar and differentiating features that focus on object key points or prominent parts of the objects, while the regression task is dedicated to accurately locating the boundary features. The two tasks are decoupled and trained independently. This may lead to inconsistent predictions because the scores obtained from the classification branch are utilized as the confidence values of the predicted bounding boxes in the inference stage. The inconsistency between regression and classification is visualized as shown in Fig. 1. There is a predicted box with a relatively high classification score in Fig. 1(a), but it can be seen that the intersection over union (IoU) between the predicted box and the ground-truth box is low. Fig. 1(b) obtains a predicted box that highly overlaps with the ground-truth box but its classification score is low. What we hope to get is high-quality predicted boxes with high IoU and classification scores, not the type of predicted boxes in Fig. 1. The inconsistency hinders the improvement of detection accuracy.

Due to the different focuses of the classification and regression tasks, decoupling the two tasks while using similar

Manuscript received 24 March 2024; revised 16 May 2024; accepted 4 June 2024. Date of publication 11 June 2024; date of current version 20 June 2024. This work was supported in part by the Key Research and Development Program of Shaanxi Province, China, under Grant 2023-YBGY-205 and Grant 2024GXYBXM-039; in part by the Natural Science Basic Research Program of Shaanxi, China, under Grant 2024JC-ZDXM-40; and in part by the Innovation Capability Support Program of Shaanxi, China, under Grant 2023-CX-TD-08. *(Corresponding author: Gang Liu.)*

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Digital Object Identifier 10.1109/TGRS.2024.3412167

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(a) (b)

Fig. 1. Comparison of predicted boxes with different classification scores. Ground-truth boxes are drawn in red and predicted ones in green. (a) Predicted box with a high classification score corresponds to a low IoU. (b) Predicted box characterized by a high IoU is associated with a low classification score.

network structures may lead to ineffective learning of the key features. For example, in classification tasks, the same type of objects in aerial images may have multiple orientations. The design of traditional visual convolution feature extraction largely overlooks rotation-invariant features. Many methods perform data augmentation by rotating training samples at different angles to achieve rotational invariance [23]. However, it is difficult for conventional CNN to introduce rotation-invariant information only through these data-augmented training samples, thereby limiting the generalization ability of the network. In addition, for regression tasks, most CNNs applied to images use 2-D convolution to extract features, paying more attention to the features in nearby regions. Objects with large aspect ratios are extremely common in aerial images, and they should give greater consideration to the characteristics of the single direction. Ordinary 2-D convolution has limitations in feature extraction of such objects and lacks feature fusion in other receptive fields.

To address the aforementioned problems, we propose a Task Integration based on a Rotated convolution kernel Network (TIR-Net) in the head structure of the detector, consisting of three modules: selective rotation of the kernel (SRK), regression feature refinement (RFR), and task integration (TI). The proposed modules can act as plug-and-play components at the corresponding position in one-stage methods conveniently. An SRK module selectively rotates the traditional convolution kernel at multiple angles in the channel dimension to extract rotation-invariant features. An RFR module extracts features focused on the single direction through 1-D convolution to improve the recognizability of objects with large aspect ratios in both horizontal and vertical orientations. A TI module is designed to integrate the feature knowledge of the regression and classification tasks, alleviating the inconsistency between them and further improving the quality of predicted bounding boxes. Sufficient experiments demonstrate a significant improvement in detection performance applied in one-stage methods equipped with our new head structure. Our work is summarized as follows.

1) The SRK rotates the convolution kernels at different angles in the channel dimension, which is embedded into the classification task head. This module is capable of extracting rotation-invariant features effectively.
2) We introduce RFR, with greater attention on feature extraction of large aspect ratio objects in both horizontal and vertical directions to improve their perceptibility.
3) TI is introduced to integrate features extracted from classification and regression. This fusion enhances feature quality, effectively alleviating the inconsistency between regression and classification tasks.

The remainder of this article is structured as follows. Section II introduces related work. Section III elaborates on our TIR-Net. Extensive experiments and analysis are presented in Section IV. Finally, the conclusions of our work are drawn in Section V.

II. Related Work

A. Oriented Object Detection

In recent years, various detectors have been introduced for object detection employing horizontal bounding boxes (HBBs) [20], [21], [22], [24]. Oriented object detectors recognize and classify objects using oriented bounding boxes (OBBs), offering more precise orientations for accurate object localization and classification. Xu et al. [25] regressed the length ratios of the four anchor edges, each corresponding to the relevant sliding offset. However, the corresponding RoI feature remains horizontal, leading to misalignment issues. R^3Det [9] achieves feature refinement and alignment by reencoding position information into feature points through pixelwise feature interpolation and introduces a novel loss to address the nondifferentiability issue of SkewIoU. S^2ANet [10] generates orientation-sensitive features and high-quality anchors through the inventive aligned convolution and rotation filter. EIA-PVT [26] significantly improves detection efficiency and accuracy through adaptive multigrained routing mechanisms and angle tokenization technology. CoF-Net [27] effectively improves feature representation and training sample selection through coarse-to-fine feature adaptation (CoF-FA) and sample allocation (CoF-SA) strategies. Various object representations and loss functions better suited for these complex objects have been extensively researched. For example, CSL [28] turns the prediction for the angles of bounding boxes into the classification problem to address the issue of discontinuous boundaries. GWD [29], KLD [30], and G-rep [31] represent anchors with arbitrary orientations as Gaussian distributions and use the corresponding loss function to approximate IoU loss. Moreover, FCOS [32] and FoveaBox [33] locate objects of interest not based on anchor boxes but through the distances from point to boundaries and the mapping relationship between points and coordinates. Cornernet [34] and Object as points [35] work by identifying and grouping multiple key points of the bounding boxes. There are some other methods to achieve precise angles by regressing point sets rather than OBBs, such as Reppoints [13]. O-DNet [36] uses pairs of midlines to represent oriented objects, enabling simpler and more efficient detection in an anchor-free manner.

B. Rotation-Invariant Networks

Rotation invariance indicates that the extracted feature maps of the original image or object remain consistent under any rotation transformation. Given the arbitrary orientation of objects in aerial images, it is essential to enhance detection accuracy by rotation-invariant features. With the continuous evolution of CNN, STN [37] introduces a new learnable spatial transformation module that manipulates the feature map

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using radial and spatial transformations to achieve rotational invariance. G-CNN [38] offers a natural generalization of CNNs, using filters to transform input features so that it can group objects from the same class, ignoring the effect of orientation as much as possible. The extraction of rotation-invariant features is one of the most fundamental ways to improve feature quality and detection accuracy essentially. Lately, numerous meticulously designed detectors have been introduced for oriented object detection from different perspectives, achieving improved results on challenging aerial images. The main tenet of the RoI transformer [11] is applying spatial transformations to the RoI, leveraging the model's geometric transformations to effectively mitigate misalignment between regional features and objects. ReDet [12] combines the rotation-equivariant network [39] with the detector and proposes rotation-invariant RoI alignment to extract high-quality and rotation-invariant features. RINet [40] achieves refined orientation-aware detection through a flexible multibranch online detector, allowing for natural rotation perception of directional objects. Pu et al. [41] proposed an adaptive rotation convolution (ARC) module, extracting features of objects with varying orientations in images. This module adaptively predicts angles to rotate the convolution kernel to handle orientation changes in different images. Mei et al. [23] introduced a new cyclic polar coordinate convolution layer (CPCCL) that converts rotational changes into translational changes by leveraging polar coordinate transformations. Mo and Zhao [42] designed a circular sampling area based on a new coordinate system, which naturally remains unaffected by continuous rotations around the center. FPNFormer [43] employs a window transformer to emphasize part of features and utilizes a dual-stream cross-scale transformer decoder to ensure both rotation invariance and equivariance. While numerous studies have made strides in rotation-invariant feature extraction, most of them focus on the backbone network of the detector, which will lead to reduced applicability of the model and higher training costs. In addition, some methods use deformable convolutional networks [44] to extract features with rotational invariance. Given these issues, we introduce the SRK module in the head network, which involves interpolating the original weights of a single-group convolutional kernel to replace the weights at discrete positions. SRK simulates the effect of rotating the convolutional kernel to extract rotation-invariant features.

C. Inconsistency Between Regression and Classification

Most one-stage detectors decouple the two tasks in the head network so that classification and regression are trained independently without sharing the network structure. However, the predicted scores from the classification branch are used to rank the confidence of predictions from the regression branch. This leads to inconsistencies between regression and classification. The inconsistency is more serious in oriented object detection, since numerous objects exhibit substantial aspect ratios in aerial images. To address this issue, DCR [47] situates an independent classification network parallel to the regression one. Then, the hard false sample is selected, and another classifier is trained to refine the classification results. Song et al. [46] introduced a straightforward method called task-aware spatial disentanglement (TSD) to address conflicting tasks. By employing task-aware proposal estimation and a detection head, it generates task-specific feature representations, mitigating the misalignment between classification and localization. RefineDet++ [48] filters out negative anchors to refine anchors and utilizes aligned convolution to adjust anchor positions, further improving localization accuracy. Wu et al. [49] used fully connected layers for classification instead of convolutional layers, the regression head remains unchanged, and the fully connected layers have higher spatial sensitivity. CFC-Net [50] decouples the classification and regression features and then constructs feature pyramids adapted to the respective tasks of classification and regression through the polarization attention module (PAM) so that it can obtain robust critical features. Feng et al. [51] created a new head structure to enhance interaction between two tasks and explicitly narrow the misalignment of the predictions through the designed assignment strategy and inconsistency loss. DAL [52] utilizes a freshly defined matching criterion to assess the possibility that the anchor contains an object of interest, and the anchors are dynamically selected and efficiently assigned labels. Double-Head R-CNN [53] introduces a decoupled sampling approach to choose proposals with sensitivity specific to the two tasks. This decoupled design enhances translation sensitivity and consistency in a straightforward and efficient manner. TIOE-Det [54] uses PHA labels to integrate posterior localization into classification and calculates balanced alignment loss to resolve the imbalanced regression loss. Differing from the aforementioned approaches, our method integrates features before outputting classification and regression results to alleviate biases between them, rather than focusing on the initial decoupled structure of the detection head.

III. Method

We enable the representative one-stage network RetinaNet [22] for oriented object detection and choose it as a baseline. The main innovation is found in the detection head module. SRK is elaborated upon in Section III-A. The architecture of RFR and TI are presented in Sections III-B and III-C respectively. An overview of the architecture of our method is shown in Fig. 2.

A. Selective Rotation of the Kernel

Due to the divergent focuses of classification and regression tasks, the former task emphasizes the prominent discriminative features of objects. Different orientations of objects may cause difficulty in distinguishing these discriminative features, highlighting the significance of rotation-invariant features for classification. Hence, the rotation of convolutional kernels is applied to the classification branch. Rotating the convolution kernel is the process of sampling and calculating new weight values for the rotated coordinates under the original convolution kernel weight. In order to achieve the transformation operations, we propose a method of selective rotation of the

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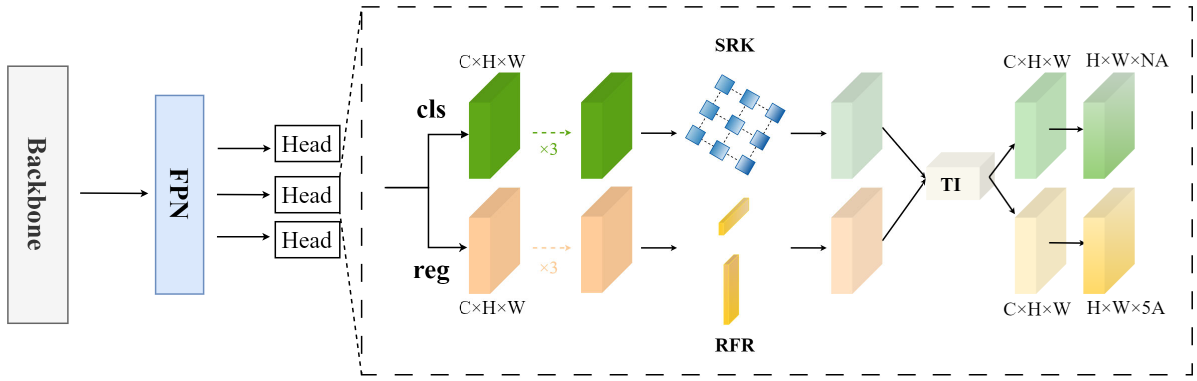


Fig. 2. Overall architecture of our method based on RetinaNet. It is made up of a backbone network, FPN [45], and two head networks. TIR-Net is within the head networks. Classification (**cls**) denotes the classification branch, and regression (**reg**) denotes the regression branch. Both the classification and regression branches start with four cascaded convolutional layers in the head. N and A represent the number of categories and anchors. The *SRK* and *RFR* modules refine the features of the classification and regression branches from different perspectives, respectively. The *TI* module aligns the features of the two branches. The final detection results are generated through the processed features.

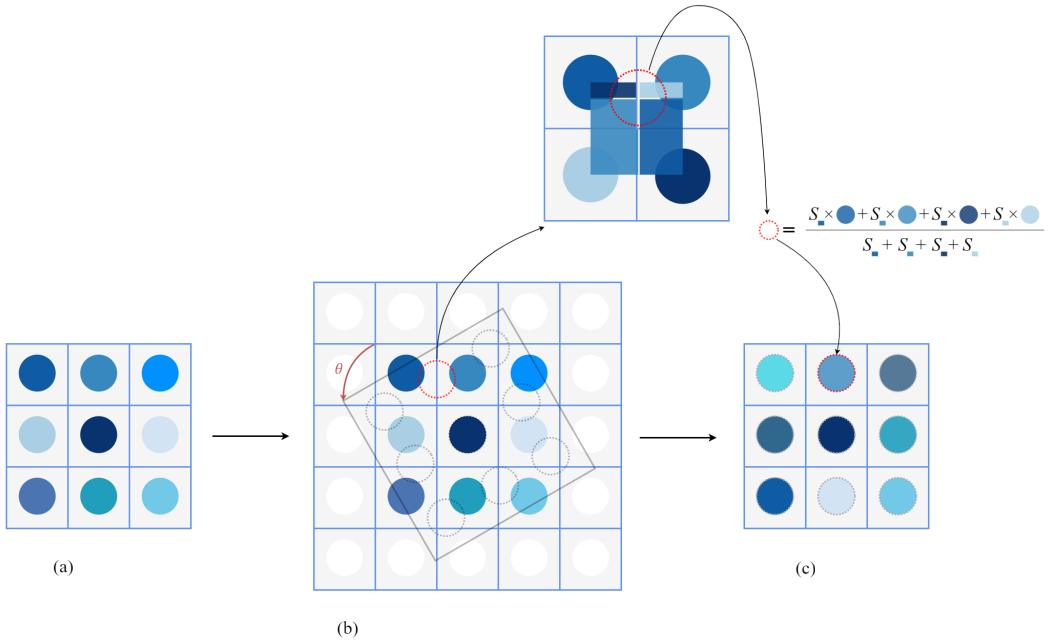


Fig. 3. Detailed description of rotating single convolution kernel. For a set of convolution kernels $K \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times k \times k}$, just repeat the process in the figure, only θ is different. The weights of convolutional kernels are represented by colored circles. The white circles in (b) indicate padding with the weight of 0, and then, the convolutional kernel in (a) is rotated counterclockwise by θ around the center point to compute the positions of the convolutional kernel weights. The specific interpolation calculation for a weight (red dashed circle) is at the top of the figure, where S denotes the area of the corresponding colored rectangle. This process is repeated for each weight to obtain the convolutional kernel in (c). It is worth noting that the weight at the center point remains unchanged.

kernel called SRK. It is worth noting that our method employs new convolution kernel weights without modifying their sampling location. First, we describe how a single convolutional kernel performs the rotation computation as shown in Fig. 3. Taking a single convolution kernel as an example (the shape of the kernel here is 3×3), the convolution kernel Kernel'_s after rotating θ can be expressed as follows:

$$\begin{aligned} \text{Kernel}'_s &= \text{Rotate}(\text{Kernel}_s, \theta) \\ &= \text{Bilinear}(\text{pad}(\text{Kernel}_s), R_c(P, \theta)) \end{aligned} \tag{1}$$

where θ is the angle at which Kernel_s is to be rotated and the convolution kernel is padding with 0 to a length of 1 by $\text{pad}(\cdot)$ to compute the weights beyond the original region after the rotation. $P = \{p_i \mid (x, y)\}$, $x, y \in \{0, 1, 2\}$ denotes the position of the initial value of the convolution kernel Kernel_s. $R_c(P, \theta)$ denotes the operation of rotating P counterclockwise by θ around the center of Kernel_s to get the rotated positions. Then, the bilinear interpolation operation, denoted by $\text{Bilinear}(\cdot, \cdot)$, is used to calculate the new weights of the kernel corresponding to the rotated positions. The coordinate axes are established with the upper left vertex as the origin, and the positive directions are oriented to the right and down.

The above describes in detail how to perform rotation calculation on a single convolution kernel. The following explains how to carry out similar operations on the convolution kernels $K \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times k \times k}$, where C_{out} and C_{in} denote the number of output channels and input channels, respectively, and k is the size of the kernels. Set the rotation angle

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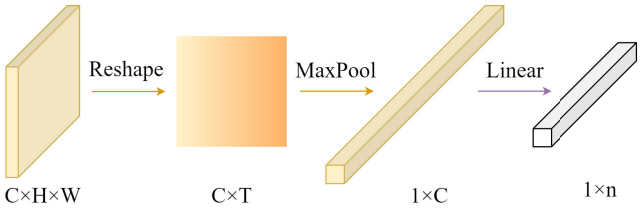


Fig. 4. Depiction of the calculation process of $\mathcal{W}$.

$\theta = [\theta_1, \theta_2, \ldots, \theta_n]$ required for the convolution kernels K, and the counterclockwise direction is defined as positive. The weights $\mathcal{W} = [w_1, w_2, \ldots, w_n]$ and θ have the same length and correspond to one to one. Based on $\mathcal{W}$, the convolution kernels K can be sliced into n small convolution kernels $K_i \in \mathbb{R}^{C_i \times C_{\text{in}} \times k \times k}$, where $C_i = C_{\text{out}} \times w_i$ and $\sum_{i=1}^{n} C_i = C_{\text{out}}$. The angle θ is set within the range of $[-90, 90)$ empirically because of the definition D_{le} of the bounding boxes. The initial sampling interval of the angle is set as follows:

$$\Delta\theta = \theta_i - \theta_{i-1} = \frac{\pi}{n}, \quad i = 2, 3, \ldots, n. \tag{2}$$

After the sampling, an increasing sequence of angles is obtained. In order to rotate the convolution kernel at different positions and angles, the angle θ is randomly disrupted. Then, the input features $x \in \mathbb{R}^{C \times H \times W}$ are reshaped into $x' \in \mathbb{R}^{C \times T}$, where $T = H \times W$. $\mathcal{W}$ is represented by x'

$$\mathcal{W} = \text{Linear}(\text{MaxPool}(x')) \tag{3}$$

where MaxPool$(\cdot)$ represents the maximum pooling layer and Linear$(\cdot)$ denotes the linear layer. The calculation process of $\mathcal{W}$ is given in Fig. 4. Then, the sliced small convolution kernels K_i are rotated by the corresponding angle

$$K_i' = \text{Rotate}(K_i, \theta_i) \tag{4}$$

where θ_i denotes the rotation angle for K_i and K_i' represents the small convolution kernels after rotating θ_i. Repeat this process for each K_i and get the final rotated convolution kernels K' and then perform a 2-D convolution on the input features x

$$F_{\text{cls}} = \text{Conv}\left(x, K'\right) \tag{5}$$

where Conv(x, K') denotes convolving the input feature x by the convolution kernels K'.

The overall algorithm of the SRK module is shown in Algorithm 1.

Algorithm 1 Selective Rotation of Kernel

Input: Traditional convolution kernels $K \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times k \times k}$.
Features $x \in \mathbb{R}^{C \times H \times W}$
Output: Rotated convolution kernels $K' \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times k \times k}$.
Initialization: $\theta = [\theta_1, \theta_2, \ldots, \theta_n]$.
$W = f(x)$, $W = [w_1, w_2, \ldots, w_n]$;
for $i \leq n$ **do**
  if $C_{\text{out}} \times w_i \geq 1$ **then**
    $K_i = [C_{\text{out}} \times w_i, C_{\text{in}}, k, k]$;
  end if
  $K_i' = Rotate(K_i, \theta_i)$;
end for
return Rotated convolution kernels $K' \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times k \times k}$.

B. Regression Feature Refinement

The regression task needs to retain the spatial information of the feature map to strengthen the ability to perceive the location information. However, relying solely on 3 × 3 convolution kernels somewhat constrains the performance of object detection. In particular, aerial images are taken from high altitudes, so there are more objects with large aspect ratios. The quality of features for slender objects extracted by the 3 × 3 kernels of traditional convolution is not good, arising from the kernels' narrow focus on the feature information within neighboring areas. So simple 1-D convolution is used instead of traditional 2-D convolution to enhance the features in horizontal and vertical directions by RFR in Fig. 5.

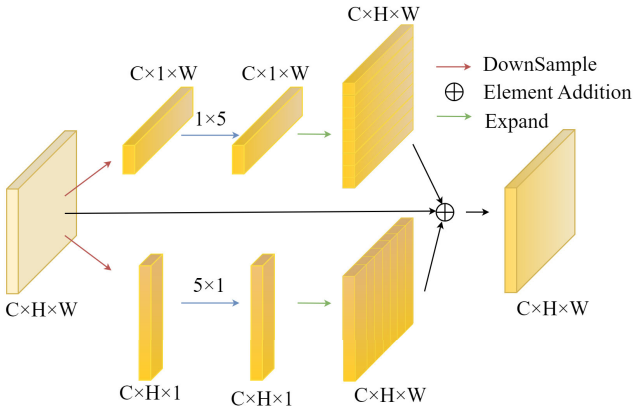


Fig. 5. RFR module. 1 × 5 and 5 × 1 represent the convolution layer with 1 × 5 and 5 × 1 kernels.

For input regression features $F_{\text{reg}} \in \mathbb{R}^{C \times H \times W}$, downsampling operations are performed on them

$$F_{\text{reg}}^{d} = \text{DownSample}\left(F_{\text{reg}}, d\right) \tag{6}$$

where the DownSample(F_{reg}, d) function denotes the sample value taken from the dth dimension of the features. F_{reg}^{h} and F_{reg}^{w} are obtained here with d being chosen as 2 and 3, respectively. Then, 1 × 5 and 5 × 1 1-D convolution operations are performed on them

$$\begin{aligned} F_{\text{reg}}^{h\,*} &= \text{Expand}\left(\text{Conv}_{1\times5}\left(F_{\text{reg}}^{h}\right)\right) \\ F_{\text{reg}}^{w\,*} &= \text{Expand}\left(\text{Conv}_{5\times1}\left(F_{\text{reg}}^{w}\right)\right) \end{aligned} \tag{7}$$

where Expand$(\cdot)$ indicates expansion in the corresponding dimension and $F_{\text{reg}}^{h\,*}, F_{\text{reg}}^{w\,*} \in \mathbb{R}^{C \times H \times W}$ represent features that focus more on the horizontal and vertical directions. The final feature fusion is defined as follows:

$$F_{\text{reg}} = F_{\text{reg}} + \lambda_1 \cdot F_{\text{reg}}^{h\,*} + \lambda_2 \cdot F_{\text{reg}}^{w\,*} \tag{8}$$

where λ_1 and λ_2 denote the learnable weight parameters. These two parameters give weights to features that focus on horizontal and vertical features. Because the proportion of the number of horizontal or vertical objects in different images is different, they are set to be learnable to find the optimal

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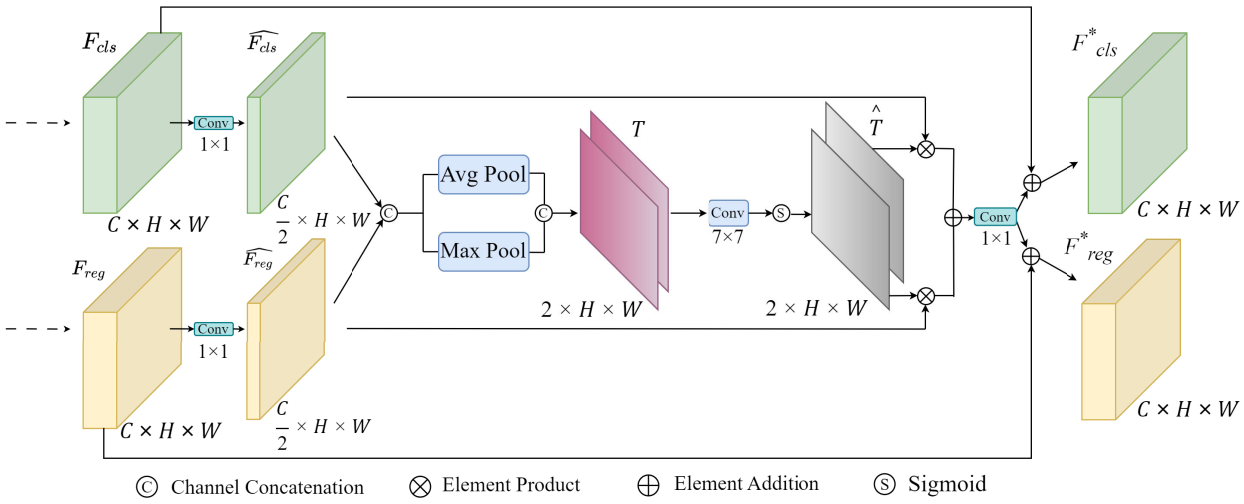


Fig. 6. Conceptual illustration of TI module.

value to fit the model. F_{reg} is used for subsequent regression information prediction.

C. Task Integration

To mitigate the inconsistency between the classification and regression task heads, we propose the TI module to further extract features from both task heads and fuse them with the original task features. The purpose of using channel-based operations instead of spatial-based operations here is to better preserve the original spatial features. First, the features obtained by convolving the regression and classification tasks are joined

$$\widehat{F_{\text{cls}}} = \text{Conv}_{1\times1}(F_{\text{cls}}), \quad \widehat{F_{\text{reg}}} = \text{Conv}_{1\times1}\left(F_{\text{reg}}\right) \tag{9}$$

$$F = \text{Concat}\left(\widehat{F_{\text{cls}}}, \widehat{F_{\text{reg}}}\right). \tag{10}$$

Note that $\text{Conv}_{1\times1}(\cdot)$ and $\text{Concat}(\cdot)$ are applied on the channel dimension. $\text{Conv}_{1\times1}(\cdot)$ denotes a change in the number of channels, and here, it refers to halving the number of channels from the original. $\text{Concat}(\cdot)$ represents the features of concatenated. F represents the features after concatenation along the channel dimension. The spatial relations are then efficiently extracted by performing average and maximum pooling in the channel dimension [denoted as $\text{AvgPool}(\cdot)$ and $\text{MaxPool}(\cdot)$] to F

$$F_{\text{avg}} = \text{AvgPool}(F), \quad F_{\text{max}} = \text{MaxPool}(F) \tag{11}$$

where F_{avg} and F_{max} are the features after average and max pooling processing. To carry out information interaction between different pooling feature descriptors, features are concatenated. The spatial attention map T is as follows:

$$T = \text{Conv}_{7\times7}(\text{Concat}(F_{\text{avg}}, F_{\text{max}})) \tag{12}$$

where $\text{Conv}_{7\times7}(\cdot)$ represents convolution operation with kernel size of 7. The spatial selection masks $\hat{T}$ are defined as follows:

$$\hat{T} = \sigma(\text{Conv}_{1\times1}(T)) \tag{13}$$

where $\sigma(\cdot)$ denotes the sigmoid function and $\text{Conv}_{1\times1}(\cdot)$ is the same as in 9, but here the number of channels is doubled.

TABLE I
PERFORMANCE EVALUATION OF COMPONENTS IN OUR METHOD

	SRK	RFR	TI	mAP_{50}
RetinaNet	-	-	-	68.42
	√			69.55 (+1.13)
		√		69.01 (+0.59)
			√	69.34 (+0.92)
	√	√		69.67 (+1.25)
		√	√	69.41 (+0.99)
	√		√	69.83 (+1.41)
	√	√	√	**70.16 (+1.74)**

Then, the features of different task branches are multiplied by the spatial selection mask and then added

$$F_{\text{fus}} = \hat{T}_1 \cdot \widehat{F_{\text{cls}}} + \hat{T}_2 \cdot \widehat{F_{\text{reg}}}. \tag{14}$$

The final outputs are the sum of the initial features and the fused features multiplied by the corresponding weights

$$\begin{aligned} F^*_{\text{cls}} &= F_{\text{cls}} + \alpha \cdot F_{\text{fus}} \\ F^*_{\text{reg}} &= F_{\text{reg}} + \beta \cdot F_{\text{fus}} \end{aligned} \tag{15}$$

where α and β are learnable weight parameters. Here, the initial values of α and β are all set to 0 to reduce the impact of the initial values on the model. As the model is trained, their values are constantly changing to find the optimal value. The obtained F^*_{cls} and F^*_{reg} are used for subsequent prediction of classification scores and localization, respectively. Fig. 6 shows the specific conceptual representation of the TI module, where the adaptive collection of features for different tasks is visualized.

IV. EXPERIMENTS

A. Datasets

DOTA-v1.0 [55] is a comprehensive dataset suitable for oriented object detection, featuring 2806 images and 188 282 instances. The dataset encompasses 15 object classes, including plane (PL), baseball diamond (BD), bridge (BR), ground

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TABLE II
EXPERIMENTAL RESULTS ON THE HRSC2016 DATASET

Method	Head	mAP_{50}	mAP_{75}	$mAP_{50:95}$
RetinaNet	-	84.80	58.10	52.06
	Ours	84.20	61.00	**53.75 (+1.69)**
S^2ANet	-	89.70	65.30	55.65
	Ours	90.20	67.70	**57.66 (+2.01)**

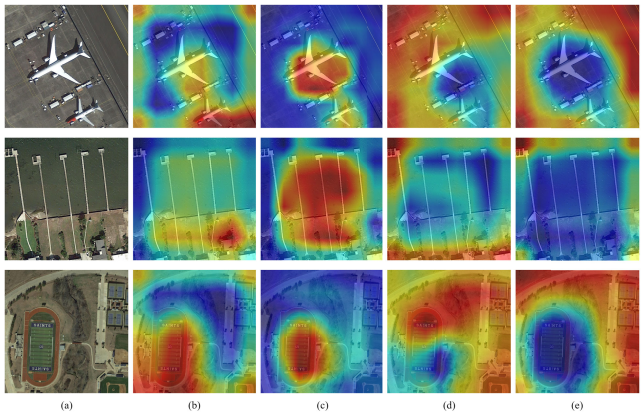


Fig. 7. Comparison of feature visualizations. The classification task and regression task are decoupled in the single-stage detector, so the features of the two tasks are visualized separately. (a) Input images. (b) Features of the classification task of baseline. (c) Features of the classification task of ours. (d) Features of the regression task of baseline. (e) Features of the regression task of ours.

TABLE III
COMPARATIVE ANALYSIS OF SRK

Baseline	Module	mAP_{50}
RetinaNet	DCN [44]	69.18
	ARC [41]	69.41
	SRK	**69.55**

TABLE IV
EVALUATION OF DIFFERENT ANGLE INTERVALS ($\Delta\theta$) IN SRK MODULE

Method	$\Delta\theta$	mAP_{50}
SRK	5	69.34
	10	**69.55**
	15	69.46
	20	69.28
	25	69.23

TABLE V
EVALUATION OF DIFFERENT CONVOLUTIONAL LAYERS IN SRK MODULE. LAYER 1, LAYER 2, AND LAYER 3 REPRESENT THE LAST THREE 3×3 CONVOLUTIONAL LAYERS FROM FRONT TO BACK IN CLS IN FIG. 2

Method	Layer1	Layer2	Layer3	mAP_{50}
SRK	√			68.50
		√		68.98
			√	**69.55**
	√	√	√	67.65

TABLE VI
ANALYSIS OF DIFFERENT SHAPES (S) OF KERNELS

Method	S	mAP_{50}
RFR	$1 \times 3, 3 \times 1$	68.96
	$1 \times 5, 5 \times 1$	**69.01**
	$1 \times 7, 7 \times 1$	68.50

TABLE VII
EVALUATION OF DIFFERENT KERNEL SIZE (k) IN TI

Method	k	mAP_{50}
TI	5	68.44
	7	**68.96**
	9	68.72

TABLE VIII
COMPARISONS WITH OTHER METHODS ON COMPUTATIONAL COST

Method	Params(M)	FLOPs(G)	FPS(frames/s)	mAP_{50}
RetinaNet	36.42	215.92	22.2	68.40
R^3Det	41.90	335.74	18.5	69.70
S^2ANet	38.60	197.62	19.0	73.91
Ours(RetinaNet-based)	37.77	192.55	20.3	70.80
Ours(R3Det-based)	43.26	312.37	17.0	71.71
Ours(S2ANet-based)	39.96	174.25	17.2	75.23

track field (GTF), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer-ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC). The images in the DOTA dataset vary in size, with the largest ones up to 4000×4000 pixels. Among them, the training set, validation set, and testing set account for 1/2, 1/6, and 1/3, respectively. To facilitate training, the raw images are cropped into small patches of 1024×1024 pixels. In order not to lose objects at the edges of small patches, the overlapping pixel width between adjacent patches is set to 200. For multiscale training and testing, the raw images are initially resized at 0.5, 1.0, and 1.5 scales and then cropped into 1024×1024 patches with an overlapping pixel width of 500. By convention, all images in the training and validation sets are used for training, and the test set is employed for evaluation. The results are calculated by submitting the testing results to the official DOTA dataset evaluation server.

HRSC2016 [56] is a widely adopted dataset of ships, comprising 1061 images with sizes ranging from 300×300 to 1500×900. There are a total of 617 images for training, with the remaining images designated for testing.

FAIR1M [57] is a fine-grained large dataset of aerial images for object detection. The dataset comprises images gathered from diverse sensors and platforms, featuring spatial resolutions spanning from 0.3 to 0.8 m. The entire dataset has a total of 37 categories and contains more than 1 million instances and more than 15 000 images. The images in FAIR1M exhibit a variable width ranging from 1000 to 10 000 pixels, contributing to the diversity and complexity of the dataset. The testing

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TABLE IX
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DOTA-V1.0 DATASET. "*" INDICATES THAT MULTISCALE TRAINING/TESTING IS USED IN THE METHOD

Method	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP_{50}
DRN [61]	88.91	80.22	43.52	63.35	73.48	70.69	84.94	90.14	83.85	84.11	50.12	58.41	67.62	68.60	52.50	70.70
PIoU [62]	80.90	69.70	24.10	60.20	38.30	64.40	64.80	90.90	77.20	70.40	46.50	37.10	57.10	61.90	64.00	60.50
RetinaNet [22]	89.29	78.54	41.13	66.29	76.92	61.68	77.40	90.89	81.37	82.89	59.05	63.79	55.05	61.95	40.14	68.42
A^2S-Det [63]	89.59	77.89	46.37	56.47	75.86	74.83	86.07	90.58	81.09	83.71	50.21	60.94	65.29	69.77	50.93	70.64
DAL [52]	88.68	76.55	45.08	66.80	67.00	76.76	79.74	90.84	79.54	78.45	57.71	62.27	69.05	73.14	60.11	71.44
R^3Det [9]	89.00	75.60	46.64	67.09	76.18	73.40	79.02	90.88	78.62	84.88	59.00	61.16	63.65	62.39	37.94	69.70
FCOS [32]	89.18	71.98	47.96	61.65	79.30	73.51	85.76	90.90	81.09	84.30	59.57	62.66	62.08	69.94	49.26	71.28
RSDet [64]	**89.80**	82.90	48.60	65.20	69.50	70.10	70.20	90.50	85.60	83.40	62.50	63.90	65.60	67.20	68.00	72.20
S^2ANet [10]	89.30	80.11	50.97	73.91	78.59	77.34	86.38	**90.91**	85.14	84.84	60.45	66.94	66.78	68.55	51.65	73.91
G-rep [31]	88.89	74.62	43.92	70.24	67.26	67.26	79.80	90.87	84.46	78.47	54.59	62.60	66.67	67.98	52.16	70.59
ARS-DETR [65]	87.65	76.54	50.64	69.85	79.76	83.91	87.92	90.26	86.24	85.09	54.58	67.01	75.62	73.66	63.39	75.47
CBDA-Net [66]	89.17	85.92	50.28	65.02	77.72	82.32	87.89	90.48	86.47	85.90	66.85	66.48	67.41	71.33	62.89	75.74
PolarDet* [67]	89.65	**87.07**	48.14	70.97	78.53	80.34	87.45	90.76	85.63	86.87	61.64	70.32	71.92	73.09	67.15	76.64
TIOE-Det* [54]	89.76	85.23	56.32	76.17	80.17	85.58	88.41	90.81	85.93	87.27	68.32	70.32	68.93	78.33	68.87	78.69
ABFL* [68]	88.85	83.02	56.03	70.08	81.46	84.47	88.11	90.84	84.68	86.58	67.97	**73.19**	75.55	**82.48**	**69.86**	78.81
S^2ANet* [10]	89.28	84.11	56.95	79.21	80.18	82.93	**89.21**	90.86	84.66	87.64	71.66	68.23	78.58	78.20	65.55	79.15
FPNFormer* [43]	89.10	82.41	56.15	81.24	**82.05**	85.16	88.74	90.89	85.78	87.07	68.93	68.47	77.93	78.65	68.35	79.39
Ours(RetinaNet-based)	89.07	80.26	42.11	68.44	77.14	65.06	78.29	90.90	81.25	82.77	59.90	62.83	57.68	66.98	49.67	70.16
Ours(R^3Det-based)	89.44	77.00	47.31	72.09	77.59	75.71	85.82	90.88	81.18	84.20	58.35	62.71	65.06	66.79	41.52	71.71
Ours(FCOS-based)	88.95	74.48	48.96	59.60	80.18	80.64	87.85	90.89	83.74	85.69	62.40	63.40	67.41	70.55	45.86	72.71
Ours(S^2ANet-based)	89.26	82.42	53.57	72.23	79.15	78.87	86.96	90.87	84.25	84.76	65.43	68.50	67.60	70.35	54.22	75.23
Ours*(S^2ANet-based)	89.56	86.99	**58.63**	**82.60**	80.52	**85.64**	88.90	90.88	**86.86**	**88.20**	**72.67**	70.09	**78.61**	80.81	68.48	**80.63**

results are calculated by submitting the results to the official ISPRS evaluation server.

UAV-ROD [58] is a car detection dataset captured by UAV. It consists of 1577 images and 30 090 instances annotated with OBBs. The numbers of images for training and testing are 1150 and 427, respectively.

B. Experimental Settings

The results on every dataset are produced on the MMRotate [59] toolbox. On the HRSC2016 dataset, RetinaNet is trained for 72 epochs, while S^2ANet is trained for 36 epochs. On the other datasets, the models are all trained for 12 epochs. The AdamW optimizer [60] is employed, and the initial training learning rate of the model is set to 0.0001 to ensure more comprehensive training and better results. Other parameters remain consistent. Although there are variations in the detailed training configurations among different detectors and datasets, consistency is maintained in the backbones and neck networks across different experiments. This is done to facilitate fair comparisons by utilizing the proposed head networks. All models are trained on GeForce RTX 3090 GPU with two images per min-batch.

The mean average precision (mAP) is used to evaluate detection accuracy, referred to as mAP_{50} and mAP_{75} when the IoU thresholds are 0.5 and 0.75. Besides, we also utilize $mAP_{50:95}$ as the auxiliary evaluation metric.

C. Effectiveness of the Modules

We demonstrate the effectiveness of our approach by making a comparison between the baseline head network and our TIR-Net on two common datasets.

On DOTA-v1.0, we experiment with the effectiveness of each module using RetinaNet as the baseline network. We add individual modules (SRK, RFR, and TI), three combinations of both, and all of them to the baseline. It can be seen in Table I that regardless of the combination, the detection accuracy of the model is improved. For a single module, SRK has the largest improvement in the model, which can increase 1.13% mAP_{50}. Using all three modules simultaneously, the model reaches the optimal mAP_{50}, which is an improvement of 1.74% compared to the baseline network. For HRSC2016, we selected RetinaNet and S^2ANet as the baseline, respectively, and added all three modules to the baseline for experiments. The experimental results are shown in Table II. Our method improves $mAP_{50:95}$ by 1.69% and 2.01% respectively.

In addition, we visualized the features and compared the baseline with our method, as shown in Fig. 7. Our method demonstrates a heightened ability to concentrate on the salient features of the object for the classification task, whereas, for the regression task, it exhibits a propensity to concentrate on peripheral details. Compared to the baseline network, there has been a significant improvement. For example, when dealing with an inclined object such as a plane, our method concentrates on the wings and the fuselage to achieve rotational invariance, thereby enabling more precise angle predictions. In the case of objects with a large aspect ratio, such as harbors, our approach is capable of focusing on the entirety of the structure rather than just specific segments. Furthermore, the integration between the regression task and the classification task has led to an improvement in the consistency of predictions, which in turn has elevated the accuracy of the detection process.

These experimental results validate that the proposed modules can effectively improve the performance of the detection network architecture on the task of oriented object detection. Our method effectively improves the generalization ability of the oriented object detection network from the results.

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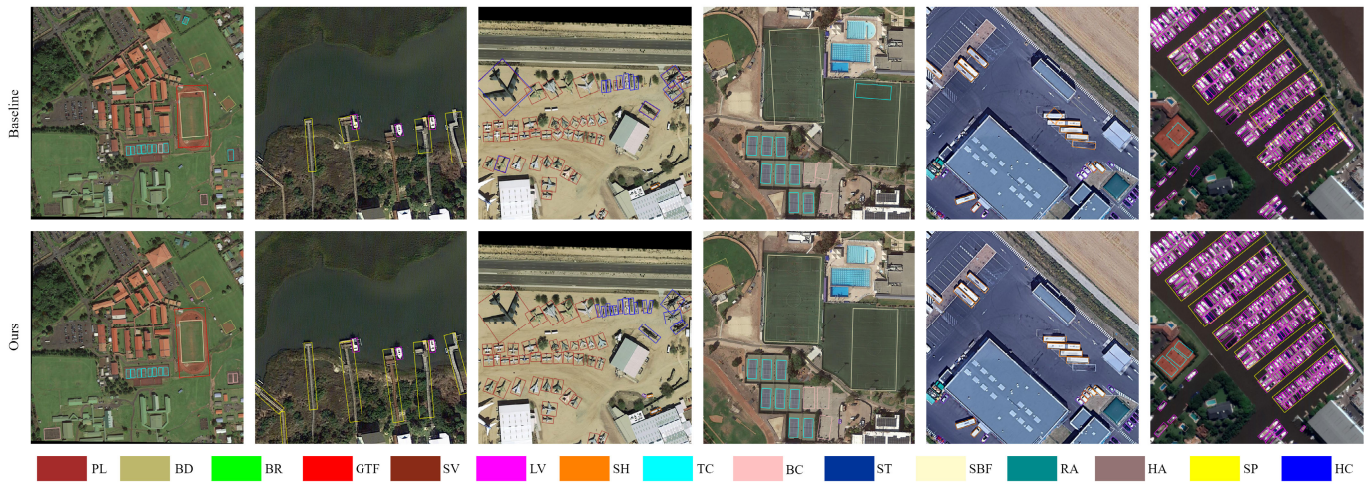


Fig. 8. Visualization of detections on DOTA-v1.0 dataset. For different detection results per image, the top image represents the result of the baseline method (S^2A-Net), while the bottom one is generated by our TIR-Net. Zoomed-in view for best view.

TABLE X
COMPARISONS WITH DIFFERENT METHODS ON HRSC2016 DATASET

Method	mAP_{50}	mAP_{75}	$mAP_{50:95}$
RetinaNet [22]	84.80	58.10	52.06
R^3Det [9]	88.52	60.40	53.42
PSC [69]	85.65	61.30	54.14
S^2ANet [10]	89.70	65.30	55.65
GWD [29]	89.43	65.88	56.07
Ours	**90.20**	**67.70**	**57.66**

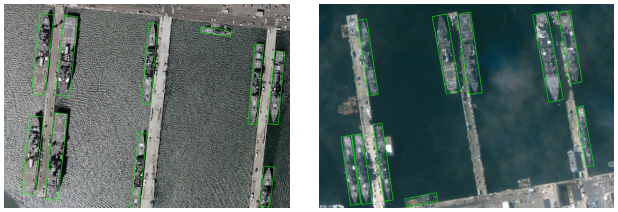


Fig. 9. Detection results on HRSC2016 dataset.

D. Ablation Studies

In this section, ablation experiments for parameter settings and replacement strategies within each module are conducted on the DOTA-v1.0 testing set to evaluate the results. Note that RetinaNet is employed as the baseline method, with the original ResNet50 + FPN serving as the backbone.

1) Comparative Analysis of SRK: In order to confirm the superiority of the SRK module, we compare it with several feature extraction modules for rotation invariance. Both ARC [41] and DCN [44] are only used in the head network. Except for the specific methods that are inconsistent, the rest of the settings are consistent with SRK, including experimental settings and the additional position of the modules. The results are shown in Table III; the accuracy of SRK is higher than the other two modules. SRK reconstructs the convolution kernel based on the output channel, which is more suitable for further refinement of features in the head network.

2) Angle Interval ($\Delta\theta$) Setting: Analytical experiments are conducted on SRK to find the optimal angle interval. The experimental results are shown in Table IV. When the angle interval ($\Delta\theta$) is set to 5, the mAP_{50} reaches 69.34%, while a setting of 10 achieves 69.55%, which is the best performance among all the intervals. The mAP_{50} gradually decreases with increasing intervals when $\Delta\theta$ is greater than 10. It is worth noting that when the weight corresponding to an angle is relatively small ($w_i \times C < 1$, where C represents the number of output channels of the kernels), we ignore this angle when partitioning the convolutional kernel because its impact on weight reconstruction is negligible. When the angle interval is set relatively small, a large number of angles are included, but it cannot be guaranteed that each angle will participate in the reconstruction of the kernels. It may achieve the same effect as when the interval is large. Therefore, we conduct ablation experiments to find a balance to optimize the model, namely, when $\Delta\theta$ is set to 10.

3) Replacement Strategy for SRK: Since SRK is an adjustment to the convolution kernel, it can be applied to all convolutional layers. Since its main function is to extract rotation-invariant features, the replacement strategy is mainly applied to the last three convolutional layers of the classification branch. The first layer of convolution remains unchanged to retain the original feature information extracted by the backbone network. The convolution layers of different locations at the classification branch are further replaced, and the results are presented in Table V. From the results, the last layer of 3×3 convolution is replaced with the proposed SRK module, resulting in 1.13% improvement in the mAP_{50} metric over the baseline model, which is the best. When replacing other layers or multiple layers, the accuracy decreases slightly. So the last 3×3 convolutional layer in the classification branch is replaced by the SRK module, and the other convolutional layers remain unchanged.

4) Shape of the Kernels for RFR: Experiments are conducted using only 1-D convolution kernels for width and height convolutions, and different shapes of 1-D convolution kernels exhibit varying abilities to extract information. As demonstrated in Table VI, when modifying the shape of 1-D convolutional kernels, changes in the model accuracy are

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TABLE XI

PERFORMANCE COMPARISON WITH STATE-OF-THE-ART METHODS ON THE FAIR1MV1.0 DATASET. "*" INDICATES THAT MULTISCALE TRAINING/TESTING IS USED IN THE METHOD. EACH ROW REPRESENTS THE AP OF THE CORRESPONDING CATEGORY

	FCOS [32]	RetinaNet [22]	CHODNet [57]	Faster RCNN [16]	CFC-Net [50]	RoI Trans. [11]	TIOE-Det* [54]	Ours*
Boeing 737	10.34	35.01	40.23	36.05	30.89	35.84	41.65	**42.72**
Boeing 747	43.54	83.72	53.39	85.19	83.87	82.74	85.39	**87.35**
Boeing 777	5.96	12.64	**29.65**	12.45	10.72	12.81	17.53	12.28
Boeing 787	13.67	36.68	54.04	45.35	38.60	43.90	48.53	**59.33**
COMAC C919	0.00	1.44	0.02	15.45	5.67	15.77	**24.32**	20.40
Airbus A220	11.71	45.44	46.26	**49.50**	42.44	48.68	45.92	45.59
Airbus A321	3.95	64.95	43.12	63.16	50.68	67.35	70.21	**70.61**
Airbus A330	15.03	58.52	27.61	65.89	55.13	65.56	63.09	**72.11**
Airbus A350	14.20	71.45	20.89	62.69	59.20	62.92	77.21	**78.56**
COMAC ARJ21	13.75	3.60	0.01	31.25	5.30	33.60	**45.32**	44.45
Passenger Ship	10.65	3.83	12.31	6.24	7.19	**15.20**	12.96	13.12
Motorboat	46.21	22.03	29.34	44.37	63.38	58.04	65.39	**68.95**
Fishing Boat	9.59	2.12	1.71	3.71	8.72	9.37	10.29	**12.10**
Tugboat	19.81	13.34	17.77	26.05	19.70	30.17	29.85	**41.10**
Engineering Ship	13.24	9.11	**17.72**	6.88	7.67	10.87	11.21	14.10
Liquid Cargo Ship	12.92	4.37	16.78	9.50	21.23	19.28	24.00	**25.89**
Dry Cargo Ship	35.08	14.49	27.51	17.78	30.54	33.02	36.01	**42.83**
Warship	20.75	3.81	13.68	6.37	23.21	24.90	32.35	**35.73**
Small Car	42.56	41.91	70.12	51.44	62.43	57.73	74.86	**75.63**
Bus	15.55	5.55	28.38	21.00	34.50	31.23	**53.31**	44.49
Cargo Truck	31.72	20.69	49.11	32.89	41.15	42.46	49.93	**51.04**
Dump Truck	23.90	16.54	44.02	40.04	42.18	45.26	55.78	**56.23**
Van	34.59	34.09	60.78	45.96	51.65	54.49	75.08	**75.16**
Trailer	12.14	0.33	14.48	7.82	11.41	15.54	19.62	**20.79**
Tractor	1.07	0.36	**4.96**	3.77	1.69	3.55	4.00	3.69
Excavator	7.90	0.52	17.60	9.28	10.26	12.78	16.62	**17.71**
Truck Tractor	1.09	0.01	**30.09**	1.71	0.71	2.59	2.18	2.48
Basketball Court	23.09	22.28	47.91	39.92	40.21	42.87	50.90	**54.44**
Tennis Court	74.76	78.62	82.11	76.97	79.41	78.40	83.97	**86.22**
Football Field	49.64	59.46	54.10	52.36	58.01	59.30	**65.29**	63.75
Baseball Field	82.90	86.46	69.97	87.56	84.34	86.60	85.96	**87.82**
Intersection	55.14	57.33	59.91	57.11	51.98	58.18	**63.36**	59.17
Roundabout	**26.46**	20.30	19.57	22.28	18.22	19.34	21.50	20.99
Bridge	22.79	9.89	14.38	7.75	14.31	20.76	**28.01**	26.92
mAP_{50}	23.70	27.67	32.93	33.70	34.31	38.27	43.87	**45.11**

observed to some extent. When the shape of the convolution kernel is 1×5 and 5×1, the mAP_{50} of the model is optimal at 69.01%. When the convolution kernel size is 1×7 and 7×1, some unimportant background feature information may be learned, resulting in a decrease in accuracy. We choose the 1×5 and 5×1 convolution kernel for convolution.

5) Convolutional Kernel Size for TI: We splice the classification and regression features and perform average and maximum pooling and subsequently convolve the features. The sizes k of the convolution kernels in (12) play a crucial role in generating subsequent spatial selection masks. Ablation experiments are conducted to analyze the impact of different convolution kernel sizes, as indicated in Table VII. The convolution kernel of size 7 is selected, where the model achieves optimal mAP_{50}.

6) Computational Cost: Table VIII reports the information on the computational cost of our method based on different baselines, including parameters, FLOPs, the inference speed (in frames/s), and performance (in mAP). The FLOPs are calculated with 1024×1024 image resolution. The frames/s is tested on an RTX 3090 with batch size 1. Take S^2ANet as an example, the baseline requires 38.60M Params with 197.62 GFLOPs, which can achieve 73.91% mAP and 19.0 frames/s. Our model based on S^2ANet achieves 75.23% mAP and 17.2 frames/s. Because our method adds three modules based on the baseline, the computational cost of ours is slightly higher (39.96M Params versus 38.60M Params).

E. Comparison With State-of-the-Art Methods

To prove the generalization of our TIR-Net, the performance was evaluated on other datasets. We all used S^2ANet as the baseline to achieve the best detection accuracy for each dataset. In addition, we also tried different baselines in DOTA-v1.0.

1) Results on DOTA-v1.0: As shown in Table IX, our method is compared with other advanced methods of oriented object detection in DOTA-v1.0. Our model achieves a detection accuracy of 75.23% mAP_{50} on single-scale images, outperforming most one-stage models. We also conduct experiments based on RetinaNet, R^3Det, and FCOS, showing that our TIR-Net is suitable for all one-stage detectors, whether anchor-based or anchor-free. It is worth noting that our method has excellent AP in categories such as bridge, ship, and

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Fig. 10. Visualization of detections on FAIR1M1.0 dataset.

TABLE XII
COMPARISONS WITH DIFFERENT METHODS ON THE UAV-ROD DATASET

Method	$mAP_{50:95}$	mAP_{75}	mAP_{50}
RetinaNet [22]	71.46	85.88	97.68
Faster R-CNN [16]	75.79	86.38	98.07
TS^4Net [58]	76.75	88.17	**98.10**
TIOE-Det [54]	77.93	89.64	97.89
Ours	**78.72**	**90.70**	98.03



Fig. 11. Illustrations of detections on UAV-ROD dataset.

harbor, which have large aspect ratios, as shown in Table IX. With restricted data augmentation such as multiscale cropping and random rotation of images, our method achieves 80.63% mAP_{50} in the whole dataset, achieving competitive results. Qualitative detections of the baseline method and ours on the DOTA dataset are visualized in Fig. 8. Compared with baseline methods, our method is more accurate in predicting angles and localizing objects with large aspect ratios.

2) Results on HRSC2016: Within the HRSC2016 dataset, there are numerous instances of thin and elongated ships exhibiting arbitrary orientations. A comparative analysis with other state-of-the-art methods is presented in Table X. Notably, our method surpasses existing approaches. Specifically, the detection accuracy of our method reaches the mAP of 57.66%, along with impressive performance in terms of AP_{50} (90.20%) and AP_{75} (67.70%). Some detection results are shown in Fig. 9.

3) Results on FAIR1Mv1.0: The performance results on the FAIR1Mv1.0 testing set are presented in Table XI. When utilizing multiscale data, our method excels with an mAP_{50} of 45.11%, surpassing the capabilities of FCOS, RetinaNet, and Faster R-CNN. Furthermore, our model sets a new benchmark in multiscale detection results compared with the best results previously achieved. Especially for multiple ship classes, our method achieves excellent detection accuracy. This underscores the efficacy of our approach in advancing the forefront of performance on the FAIR1Mv1.0 dataset. Visualization of some detection results is shown in Fig. 10. Our method achieves good detection results, especially in dense scenes.

4) Results on UAV-ROD: The corresponding results of UAV-ROD are detailed in Table XII. While our method exhibits a slightly lower AP_{50} compared to models like TS^4Net, it excels in enhancing high-precision detection performance. Specifically, our approach achieves an overall mAP of 78.72% and an AP_{75} of 90.70%, marking the highest performance among the models under comparison. This emphasizes the specialization of our method in achieving superior precision, especially evident in the outstanding AP_{75} metric. Illustrated in Fig. 11 are certain detection outcomes on UAV-ROD.

V. CONCLUSION

In this article, we propose a TIR-Net consisting of three modules: SRK, RFR, and TI. SRK module and RFR module refine the features of classification and regression tasks, respectively, introducing rotation-invariant features and features of large aspect ratio objects. TI module integrates the refined task features to further improve feature quality and effectively alleviate the inconsistency between regression and classification tasks. Our method achieves excellent performance on several publicly available aerial image datasets. A large number of experimental results demonstrate the effectiveness of our method.

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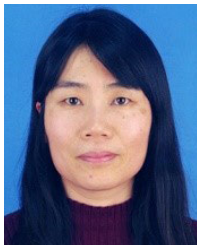
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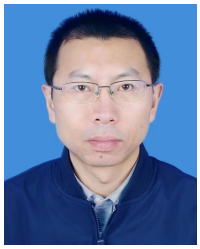
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